
CAPSTONE PROJECT

AI AGENT FOR SMART FARMING ADVICE

Presented By:

APEKSHA K J, PRESIDENCY UNIVERSITY, CSE[BLOCKCHAIN]

OUTLINE

- ☒ **Problem Statement**
- ☒ **Proposed System/Solution**
- ☒ **System Development Approach (Technology Used)**
- ☒ **Result (Output Image)**
- ☒ **Conclusion**
- ☒ **Future Scope**
- ☒ **References**

PROBLEM STATEMENT

- ⊠ Small-scale farmers often lack access to real-time, localized agricultural information in their own language. This leads to poor decisions on crop selection, pest control, and market timing. There is a need for an AI-powered assistant, using Retrieval-Augmented Generation (RAG), that provides trusted, region-specific farming advice—such as weather updates, soil conditions, crop suggestions, and mandi prices—in local languages. This will help farmers make informed decisions, reduce risk, and improve their income.

PROPOSED SOLUTION

Developing an AI Agent for Smart Farming Advice , powered by RAG (Retrieval-Augmented Generation) and deployed using IBM Cloud Lite services and IBM Granite LLM

- ☐ Accept text in regional languages
- ☐ Retrieve real-time data from trusted agricultural sources, including
 - Weather forecasts
 - Soil conditions
 - Crop recommendations
 - Pest control methods
 - Current mandi prices
- ☐ Use IBM Granite to generate intelligent, natural-language responses based on the retrieved data.
- ☐ Help farmers make informed, data-driven decisions by delivering personalized suggestions in a user-friendly interface.

SYSTEM APPROACH

The System Approach outlines the overall strategy, methodology, tools, and technologies used to develop and implement the AI Agent for Smart Farming Advice powered by Retrieval-Augmented Generation (RAG) and IBM Cloud services.

☐ System requirements:

Use of IBM Cloud Lite services / IBM Granite

☐ Libraries and services required

- watsonx.ai runtime
- API Keys
- LangGraph

RESULT

The screenshot displays the IBM watsonx Agent web interface. The browser address bar shows the URL: `datapatform.cloud.ibm.com/wx/agents/7b7018a8-3ee4-4ba6-a082-03334390378c?project_id=5019b0a3-3014-4952-bc9b-ac2de1d19e07&context=wx`. The interface is divided into two main panels: 'Build' on the left and 'Agent preview' on the right.

Build Panel:

- Model:** Set to `granite-3-3-8b-instruct`.
- Instructions:** A list of instructions for the agent, including:
 - IBM Granite model used for generating answers
 - Hosted on IBM Cloud File
 - Documentation of the system and setup steps for real life times.
- Knowledge:** A section for vector index, currently showing `crop_calendar_by_harvesting_time_1`. A note states: "If you want to add a vector index to the agent, clear the agent preview chat."
- Tools:** Options to 'Add a tool' or 'Create custom tool'. Under 'Added tools (0)', there are two tools:
 - Google search:** Retrieves information from the Internet with the Google search engine.
 - DuckDuckGo search:** (Partially visible)

Agent preview Panel:

- User Query:** "What is today's mandi rate for tomatoes?"
- Agent Response:**

The mandi rates for tomatoes vary by region in India. Here are some current rates as of today:

 - In Punjab, the mandi rate for tomatoes is approximately ₹41.7 per kg.
 - In Madhya Pradesh, it's ₹2696.73 per quintal (100 kg).
 - In Uttar Pradesh, the rate is ₹3528.92 per quintal.
 - In Haryana, tomatoes are selling at ₹4259.76 per quintal.
 - In Maharashtra, the rates are not specified in the provided sources.
 - In Delhi (NCT), the mandi rate is ₹6000 per quintal.

Please note that these rates are subject to change throughout the day as market conditions fluctuate. For the most accurate and up-to-date information, it's recommended to check with local agricultural market authorities or reliable agricultural news sources.

► How did I get this answer?
- Input Field:** A text box at the bottom with the placeholder "Type something..." and a send button.

The Windows taskbar at the bottom shows the date and time as 17:13 on 08/08/2025, along with system icons for network, volume, and battery.

Agent Lab (beta) | Docs | IBM

watsonx Agent | Smart Farming | Identity & Access Management

dataplatform.cloud.ibm.com/wx/agents/7b7018a8-3eee-4ba6-a082-03334390378c?project_id=5019b0a3-3014-4952-bc9b-ac2de1d19e07&context=wx

Incognito

IBM watsonx

Upgrade ?

Apkshita K J's Account

Dallas AK

Projects / Smart Farming Advice Agent / watsonx Agent

Share feedback

Autosaved 5:31 PM

New agent +

Deploy

Build

Model: granite-3-3-8b-instruct

Agent preview

Setup

Configuration

Framework

Architecture

Instructions

Knowledge

Tools

LangGraph

ReAct

Advanced configuration

Clear chat

Cloud Services | **IBM Cloud IaaS** (Mandatory)

Language Model | **IBM Granite** (Mandatory)

Chatbot Interface | IBM Watson Assistant

Backend Logic | IBM Cloud Functions

crop_calendar_by_harvesting_time_1

If you want to add a vector index to the agent, clear the agent preview chat.

south india

watsonx Agent 05:31 PM

In South India during August, the weather is typically hot and humid, with the monsoon season in full swing. Given these conditions, some crops that would be suitable for this time include:

1. **Rice:** Being a staple crop in South India, rice thrives in the monsoon conditions.

2. **Corn:** This crop also does well in the summer heat and can benefit from the increased rainfall.

3. **Turmeric:** A cash crop that is well-suited to the wet conditions.

4. **Green Peas:** Some varieties can be planted in the summer, especially in areas with some shade or cooler microclimates.

5. **Beans:** Certain varieties of beans can be planted in the summer, especially in areas with adequate water management.

Please note that these are general suggestions. The best crop for your specific farm would depend on detailed factors like soil pH, nutrient levels, and local pest pressures.

To give you the most accurate advice, I would need to retrieve real-time data on your local weather, soil conditions, and any relevant agricultural advisories for your region in South India.

Would you like me to fetch this information?

► How did I get this answer?

Type something...

27°C Light rain

Search

ENG IN

17:32 08.08.2025

Agent Lab (beta) | Docs | IBM | watsonx Agent | Smart farming | Identity & Access Management |

dataplatform.cloud.ibm.com/wx/agents/7b7018a8-3eee-4ba6-a082-03334390378c?project_id=5019b0a3-3014-4952-bc9b-ac2de1d19e07&context=wx

IBM watsonx

Upgrade | ? | 🔔 | Apaksha K J's Account | Dallas | AK

Projects / Smart Farming Advice Agent / watsonx Agent

Share feedback | 🟢 Autosaved 5:43 PM | 📅 | New agent + | Deploy | 🔒 | ⓘ

Build

Model: granite-3-3-8b-instruct

Setup

Configuration

Knowledge

Tools

Add a tool | Create custom tool

Added tools (1)

Document search - crop_calendar_by_harvesting_time_1

Search documents with vector indexes.

Agent preview

watsonx Agent 05:41 PM

Hi, I am Smart farming Advice Agent. How can I help you?

▶ How did I get this answer?

AK You: 05:42 PM

what is the harvesting season pl turnip crop?

watsonx Agent 05:42 PM

The harvesting season for turnip crops is typically between 2 months after planting.

▶ How did I get this answer?

AK You: 05:43 PM

what is the planting season for turnip crop?

watsonx Agent 05:43 PM

The planting season for turnip crops is typically from March to August.

▶ How did I get this answer?

Type something...

AI agent instructions x CROP CALENDAR ARRANGEMENT x

File C:\Users\akank\OneDrive\Pictures\Camera%20Roll\crop_calendar_by_harvesting_time_1.pdf

CROP CALENDAR ARRANGED BY SHORTEST HARVEST SEASON TO LONGE... 1 / 1 100% +

CROP INFORMATION ARRANGED BY HARVEST SEASON
(SHORTEST TO LONGEST)

Crop type	Harvesting season	Spacing	Planting season
Of Krop we yu save harvestem afta la 1 manis kasem 3 manis			
Radish	1 - 1.5 months AP	0.25m X 0.25m	Year round
Island cabbage	1 - 1.5 months AP	0.5m X 0.5m	Year round
Chinese cabbage	2 months AP	0.4m X 0.3m	March - September
Turnip	2 months AP	0.4m X 0.2m	March - August
Cucumber	2 months AP	2m X 1.5m	Year round
Bean	2 months AP	0.7m X 0.5m	March - August
Lettuce	2 - 3 months AP	0.4m X 0.3m	March - September
Parsley	2 - 3 months AP	0.4m X 0.2m	March - October
Carrot	2 - 3 months AP	0.4m X 0.2m	March - September
Tomato	2 - 3 months AP	0.7m X 0.6m	February - September
Capsicum	2 - 3 months AP	0.7m X 0.6m	Year round
Rock melo	2 - 3 months AP	2m X 1.5m	April - August
Water melon	2 - 3 months AP	2m X 1.5m	Year round
Of Krop we yu save harvestem la 3 manis kasem 6 manis			
Beetroot	3 months AP	0.4m X 0.2m	March - August
Irish potato	3 - 4 months AP	0.8m X 0.4m	April - September
Peanut	3 - 4 months AP	0.5m X 0.3m	April - September
English cabbage	3 - 4 months AP	0.7m X 0.6m	March - September
Kumala	3 - 5 months AP	1m X 0.5m	April - September
Egg plant	4 - 5 months AP	0.7m X 0.6m	Year round
Spring onion	4 - 6 months AP	0.4m X 0.2m	March - July
Of Krop we yu save harvestem la 6 manis iga antap			
Onion	6 - 7 months AP	0.4m X 0.2m	April

CONCLUSION

- ⊠ The proposed AI-powered Smart Farming Advice Agent, built using IBM Cloud Lite and IBM Granite, demonstrates a scalable solution for delivering real-time, localized, and multilingual agricultural assistance to small-scale farmers. By integrating Retrieval-Augmented Generation (RAG), the system ensures reliable and contextual responses using trusted data sources such as weather forecasts, soil insights, market prices, and crop recommendations. This not only bridges the knowledge gap in rural farming communities but also supports informed decision-making that increases crop yield, reduces risk, and improves income.

FUTURE SCOPE

- ❑ **Voice-based Interaction:** Integrate speech-to-text and text-to-speech for hands-free usage, especially for illiterate farmers.
- ❑ **Expanded Data Sources:** Include satellite imagery, crop disease diagnostics via image inputs, and more mandi integration.
- ❑ **Offline Capabilities:** Build lightweight mobile or edge AI versions for areas with low or no internet connectivity.
- ❑ **Blockchain Integration:** Enable traceability and transparency in the supply chain to benefit both farmers and consumers.
- ❑ **AI Personalization:** Tailor crop and input recommendations based on individual farm history and local climatic trends using AI analytics.

REFERENCES

- ⊠ IBM Watson Decision Platform for Agriculture – <https://www.ibm.com/products/watson-decision-platform>
- ⊠ IBM Granite Foundation Models – <https://www.ibm.com/products/granite>
- ⊠ Building AI Agents with RAG and Watson Discovery – <https://www.ibm.com/products/watson-discovery>

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Completion Certificate



This certificate is presented to

Apeksha K J

for the completion of

Lab: Retrieval Augmented Generation with LangChain

(ALM-COURSE_3824998)

According to the Adobe Learning Manager system of record

Completion date: 24 Jul 2025 (GMT)

Learning hours: 20 mins



THANK YOU